Physiological Characteristics of the Peripheral Bioimpedance Waveform

1. Morphological Divergence from Standard Pressure Waves

The measured Bioimpedance (BioZ) signal at the Dorsalis Pedis (APW2) often exhibits a sharper, more irregular profile than standard arterial pressure illustrations. This is driven by several biological factors:

- **Peripheral Pulse Amplification:** As the pressure wave travels from the elastic aorta (APW1) to the stiffer peripheral arteries (APW2), the peak pressure increases and the waveform narrows. This phenomenon, known as distal pulse amplification, results in a steeper systolic upstroke and a more pointed waveform at peripheral sites.
- **Vascular Resistance and Compliance:** The Dorsalis Pedis artery is surrounded by less soft tissue and is constrained by bony structures of the foot. This reduced local compliance leads to faster systolic-to-diastolic transitions, producing a sharper, V-shaped waveform.
- **Fluid Conductance vs. Pressure:** Unlike pressure-based measurements, BioZ tracks the instantaneous volume of conductive blood. Variations in hematocrit and erythrocyte orientation during high-velocity flow introduce fine physiological variations not present in pure pressure waveforms.

2. The Origin of the Secondary (Double) Peak

The presence of two distinct peaks within a single cardiac cycle in the APW2 signal is a well-established hemodynamic phenomenon:

- **Wave Reflection:** When the systolic wave reaches high-resistance arterioles and capillary beds, part of the wave reflects back. This reflected component overlaps with the forward wave, forming a secondary peak often referred to as the dicrotic wave.
- **Dicrotic Notch Significance:** The dip between the peaks corresponds to aortic valve closure. At peripheral sites, this notch becomes more pronounced due to increased temporal separation caused by propagation delay.
- **Arterial Resonance:** The arterial system behaves like a resonant structure. In compliant (healthy) arteries, wave reflections are distinct, producing a visible second peak. In stiffer arteries, these peaks merge into a broader waveform.



Figure 2: apw2

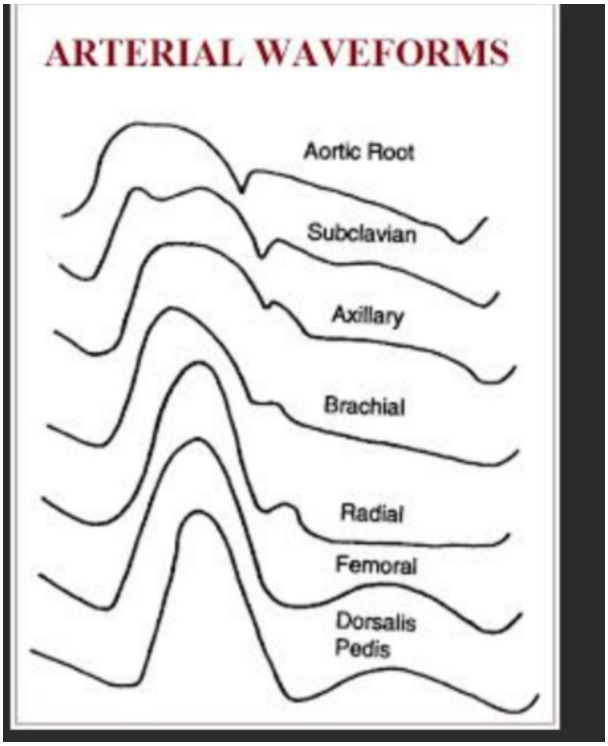


Figure 1: standard dorselis waveform , the diastolic peak not that high, diastolic notch is also deep like our waveform

3. Influence of Respiration on Bioimpedance and ECG Signals

1. Suppression of Respiratory Effects in BioZ Measurements

In this system, the primary objective is to capture blood-related bioimpedance variations to extract arterial pulse waveforms, rather than respiratory signals. Consequently, the effect of breathing is not prominently visible in the BioZ signal under normal conditions. This is achieved through both physiological and signal processing considerations:

- **Measurement Focus:** The BioZ acquisition is tuned to emphasize rapid impedance changes caused by pulsatile blood flow in arteries, which occur at the cardiac frequency range, while slower respiratory-induced variations are treated as interference.
- **High-Pass Filtering:** A high-pass filter is applied to the BioZ signal to suppress low-

frequency components associated with respiration. Since breathing typically occurs at a much lower frequency than cardiac activity, this filtering effectively attenuates respiratory modulation while preserving the arterial pulse waveform.

- **Electrode Placement Strategy:** The electrode configuration is optimized for arterial pathways (e.g., subclavian to dorsalis pedis), which enhances sensitivity to blood volume changes rather than thoracic impedance variations caused by lung expansion.

2. Residual Respiratory Influence Under Vigorous Breathing

Despite these measures, respiration can still influence the BioZ signal under conditions of deep or rapid breathing:

- **Large Thoracic Impedance Changes:** Vigorous breathing causes significant changes in lung volume and chest impedance, which can leak into the measured BioZ signal and partially saturate it.
- **Masking of Pulse Waveforms:** When respiratory variations become dominant, they can overshadow the smaller-amplitude arterial pulsations, leading to a visible reduction or distortion of pulse waveform clarity in the GUI.
- **Nonlinear Mixing Effects:** The coexistence of strong respiratory and cardiac components can introduce nonlinear interactions, further degrading the separability of the pulse waveform.

3. Respiratory Modulation Observed in ECG Signals

Unlike BioZ, the ECG signal does not directly measure impedance but reflects the electrical activity of the heart. However, respiration still indirectly modulates ECG characteristics:

- **Respiratory Sinus Arrhythmia (RSA):** During slow, deep breathing, the heart rate decreases during exhalation and increases during inhalation. This results in an increased spacing between successive R-peaks in the ECG waveform.
- **Heart Rate Variation with Breathing Rate:** Fast breathing leads to an overall increase in heart rate, reducing the interval between R-peaks, whereas slow breathing produces the opposite effect.

BP Estimation Model and Generalization Across Patients

1. BP Estimation Model Used in Code

The model estimates systolic and diastolic blood pressure as nonlinear functions of pulse transit time (PTT) using engineered features. The feature vector is defined as:

$$\mathbf{x} = \left[PTT,\ \frac{1}{PTT},\ \frac{1}{PTT^2},\ \ln(PTT) \right]$$

The trained XGBoost-based multi-output regressor learns a mapping:

$$\hat{BP} = f(\mathbf{x})$$

where $\hat{BP} = [\hat{SBP},\ \hat{DBP}]$.

Expanding this explicitly:

$$\hat{BP} = f\left(PTT,\ \frac{1}{PTT},\ \frac{1}{PTT^2},\ \ln(PTT)\right)$$

Thus, the estimated systolic and diastolic pressures are:

$$\hat{SBP} = f_1\left(PTT,\ \frac{1}{PTT},\ \frac{1}{PTT^2},\ \ln(PTT)\right)$$

$$\hat{DBP} = f_2\left(PTT,\ \frac{1}{PTT},\ \frac{1}{PTT^2},\ \ln(PTT)\right)$$

Note: The functions f_1 and f_2 are implicit nonlinear mappings learned by the ensemble of decision trees in the XGBoost model.

2. Physiological Basis of the Model

From hemodynamic theory, the relationship between BP and PTT is nonlinear and can be approximated as:

$$BP = a_0 + \sqrt{a_1 + \frac{a_2}{PTT^2}}$$

where a_0, a_1, a_2 are subject-specific parameters dependent on arterial stiffness, vessel geometry, and blood properties.

The engineered features $\frac{1}{PTT}$ and $\frac{1}{PTT^2}$ in the model are directly motivated by this physical relationship.

3. Can the Model be Used for Other Patients?

Although the model captures the nonlinear relationship between PTT and BP effectively for the given dataset, it **cannot be directly generalized to other patients without calibration**.

4. Subject-Specific Nature of PTT–BP Relationship

The limitation arises due to physiological variability across individuals. As stated in the literature:

> "The PTT-BP relationship is individual-specific and depends on the physical properties of the vessels and blood of each person. Therefore, a calibration procedure is required for each individual."

Thus, the mapping $BP = f(PTT)$ is not universal.

5. Implications for the Current Model

The trained model is effectively a **population-level model**, meaning:

- It learns an average mapping between PTT and BP across subjects.
- It may perform reasonably for similar individuals.
- It can produce significant errors for unseen patients with different vascular properties.

6. Requirement of Calibration

To ensure accurate BP estimation for a new subject, a calibration step is required:

- Measure reference BP using a cuff.
- Record corresponding PTT values.
- Fit subject-specific parameters or fine-tune the model.

Once calibrated, the model can be used for continuous cuffless BP monitoring for that individual.

7. Conclusion

PTT-based BP estimation provides a promising non-invasive approach; however, it is **not universally transferable**. Accurate prediction requires **subject-specific calibration**, without which the model may not generalize reliably to other patients.

Use of Two Right Leg Drive (RLD) Electrodes

In the implemented system, two separate analog front-end (AFE) chips (MAX30001) are used simultaneously for multi-channel bio-signal acquisition. Each AFE breakout board is designed with a standard three-electrode interface consisting of Left Arm (LA), Right Arm (RA), and Right Leg Drive (RLD), typically provided via a single jack connector.

1. Role of the RLD Electrode

The Right Leg Drive (RLD) electrode is used to improve common-mode noise rejection. It actively drives the body to a reference potential, thereby reducing power-line interference and stabilizing the measured bioelectric signals. Each AFE chip internally uses its own RLD feedback loop for optimal performance.

2. Practical Constraint in Hardware Design

Ideally, a shared RLD electrode could be used for both AFE chips, since both systems are connected to the same human body. However, in this setup, each breakout board integrates its own RLD connection through a dedicated jack.

While it is theoretically possible to connect the RLD outputs of both chips to a single electrode, doing so would require physically modifying the electrode cables (e.g., cutting and merging RLD lines). Since the original electrode connectors provided by Protocentral were not replaceable or readily available, this approach posed a significant risk of damaging the hardware.

3. Design Decision

To avoid potential hardware damage and ensure reliable operation:

- Two separate RLD electrodes were used, one for each MAX30001 AFE chip.
- No modification of the manufacturer-provided electrode cables was performed.

4. Cost Consideration

An alternative solution would have been to procure an additional Protocentral electrode interface module or accessories, which would incur additional cost (approximately INR 5000). Given

resource constraints, the safer approach of using independent RLD connections was adopted.

5. Conclusion

Although sharing a single RLD electrode is theoretically feasible, practical hardware limitations and risk considerations led to the use of two independent RLD electrodes. This ensures proper functioning of both AFE chips without compromising signal integrity or hardware reliability.